FELIX JIMENEZ

Work Authorization: U.S. Citizen

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EXPERIENCE

Research Assistant

University of Wisconsin—Madison / Texas A&M University

Dec. 2020 – May 2025

- Authored research on scalable Bayesian optimization, variational inference, and uncertainty quantification in deep learning.
- Presented work at conferences, workshops, university seminars, and seminars at tech companies and government labs.

Research Intern

Toyota Research Institute

May 2023 – Aug. 2023 Los Altos, CA

- Designed a self-supervised variational autoencoder for interpretable embeddings of EV lithium-ion cell diagnostics.
- Drafted a manuscript on developed methods; created a Python package and contributed to patent evaluation.

Research Intern

Microsoft Research

May 2022 – Aug. 2022 Cambridge, MA

- Created a dataset of CNN training loss curves and gradient statistics across diverse hyperparameters.
- Developed a Dirichlet mixture of Gaussian processes to cluster loss curves for Bayesian optimization.

Mathematical Statistician (Prev. Research Assistant)

National Institute of Standards and Technology

May 2018 – Sept. 2020 Boulder, CO

- Co-authored methodological and domain-specific work on topics such as GANs and Bayesian modeling.
- Presented research on applied machine learning at ASA conferences (JSM, QPRC) and seminars for government agencies.

SELECTED WORKS

Jimenez, F. & Katzfuss, M., "Probabilistic Skip Connections for Deterministic Uncertainty Quantification in Deep Neural Networks." *ArXiv Preprint*, 2025. [arXiv]

Jimenez, F. & Katzfuss, M., "Vecchia Gaussian Process Ensembles on Internal Representations of Deep Neural Networks." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025. [arXiv]

Cao, J., Kang, M., Jimenez, F., Sang, H., Schafer, F. & Katzfuss, M., "Variational sparse inverse Cholesky approximation for latent Gaussian processes via double Kullback-Leibler minimization." *International Conference on Machine Learning (ICML)*, 2023. [arXiv]

Jimenez, F. & Katzfuss, M., "Scalable Bayesian Optimization using Vecchia Approximations of Gaussian Processes." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023. [arXiv]

ABOUT ME

I am in the final year of my PhD in Statistics at the University of Wisconsin-Madison, working with Matthias Katzfuss. My research focuses on scalable and uncertainty-aware machine learning.

EDUCATION

Ph.D., Statistics

University of Wisconsin—Madison

2023 – 2025 Madison, WI

Texas A&M University

2020 – 2023 College Station, TX

M.S. / B.S., Applied Mathematics

University of Colorado Boulder

2014 – 2018 Boulder, CO

TECHNICAL SKILLS

Programming

- Programming: Python, Bash
- Frameworks: PyTorch, JAX (basic)
- Tools: Docker, Git, Slurm/HTCondor, Linux (Ubuntu/Arch), LaTeX
- Infrastructure: AWS (EC2, S3), GPU workflows (CUDA/PyTorch)

ML & Stats

- Bayesian optimization & Gaussian processes
- Uncertainty quantification & probabilistic modeling
- Deep learning & generative modeling

MISCELLANEOUS

Service

- Mentored undergraduate researchers at TAMU and UW-Madison.
- Served in student government at TAMU.
- Reviewer for AISTATS ('23, '25) and CVPR ('25).